

Domain and Range from Problem Situations

TEACHER KEY | Algebra II Honors | Unit 1 | Day 2 | TEKS A2.7.I, A2.2.A

Objective: I can build a function from a described situation, determine the domain the situation allows versus the domain the equation allows, and interpret both in context.

Vocabulary

Function domain — every input the **equation** accepts. **Contextual domain** — every input the **situation** allows.

Constraint — a condition the situation imposes that the algebra does not, such as "a length must be **positive**."

Average rate of change from a to $b = \frac{f(b) - f(a)}{b - a}$ — the slope between two points.

Part 1 — Build the Function First

You are given the situation, not the equation. Write the rule, then the domain.

- a. A rectangle has perimeter 40 ft. One side is x . Write its area as a function of x .

Other side = **$20 - x$** $A(x) = x(20 - x)$ Contextual domain = **$0 < x < 20$, $(0, 20)$**

- b. Why is the domain open at both ends rather than closed? **at $x = 0$ or $x = 20$ there is no rectangle**
- c. What domain does the equation $x(20 - x)$ accept on its own? **all real numbers**

Part 2 — Three Situations

Write the rule, the contextual domain, and the range. Use interval or set notation as fits.

1. A taxi charges \$4 plus \$2.50 per mile, for trips up to 12 miles.

$C(m) = 4 + 2.5m$ Domain **$[0, 12]$** Range **$[4, 34]$**

2. A club sells shirts for \$15. Each shirt costs them \$6, and the screen fee is \$90 total.

Profit $P(n) = 15n - 6n - 90 = 9n - 90$ Break-even at $n = 10$

Domain **$n = 0, 1, 2, \dots$ (whole shirts)** Discrete or continuous? **discrete**

3. A ball is thrown upward: $h(t) = -16t^2 + 48t + 4$ feet, until it lands.

Domain: from $t = 0$ until $h = 0$, about **$[0, 3.08]$** Why can t not be negative? **time starts at the throw**

The range starts at $h = 0$ and the maximum is at $t = 1.5$, giving $h = 40$, so range **$[0, 40]$**

Part 3 — Average Rate of Change

Use the taxi function $C(m) = 4 + 2.5m$ and the ball function $h(t)$ from above.

- a. Average rate of change of C from $m = 2$ to $m = 8$:

= **2.5 dollars per mile** Why is this the same for every interval? **C is linear**

- b. Average rate of change of h from $t = 0$ to $t = 1$, then from $t = 2$ to $t = 3$:

First: **32 ft/s** Second: **-32 ft/s** What does the sign change mean? **the ball is falling**

Part 4 — Going Further

1. Describe a real situation whose contextual domain is exactly the whole numbers 0 to 30.

2. Give a situation where the range is bounded but the domain is not.

Sample: **a thermostat holding a room between 68° and 72° for all time**

3. Can a contextual domain ever be LARGER than the function domain? Explain.

No — **the situation can only restrict what the equation already allows**

Exit Ticket

A square sheet of metal 24 cm on a side has a square of side x cut from each corner. Write the volume of the resulting open box and give the contextual domain, with justification.

$V(x) = x(24 - 2x)^2$ Domain **(0, 12)**

Justify the upper bound: **at $x = 12$ the side $24 - 2x$ is 0, so no box exists**